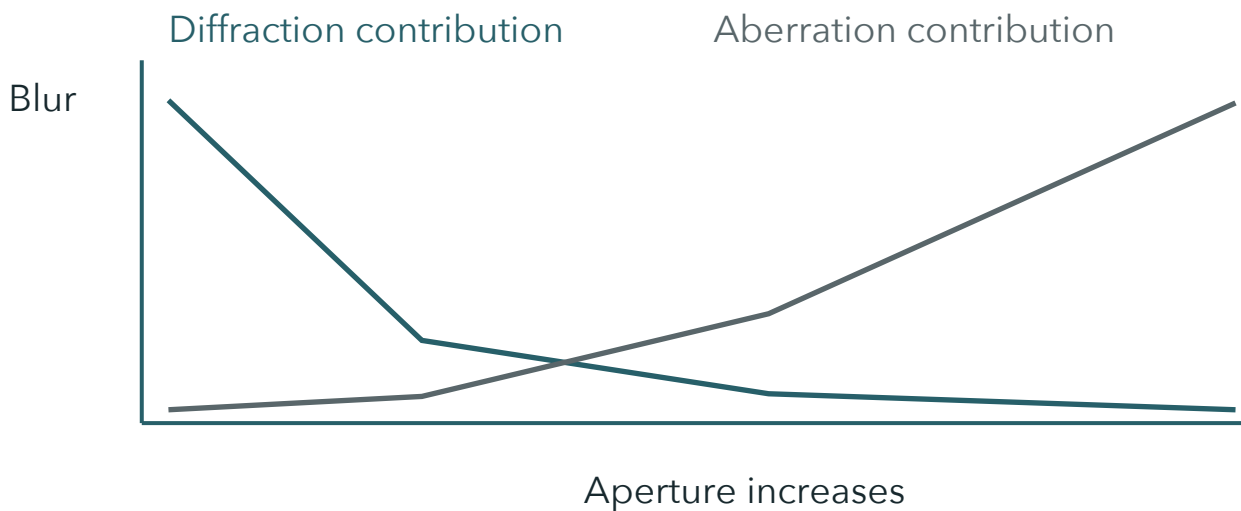


Pupil size, diffraction and optical contrast

Part 1 | The pupil sets competing optical limits

Reducing aperture can suppress aberrated rays while increasing diffraction.

Qualitative curves; no universal optimum implied



A qualitative trade-off shows greater diffraction at tiny apertures and greater aberration contribution at larger ones.

A useful extreme

An extremely small aperture does not deliver unlimited resolution. The diffraction angle increases as wavelength divided by aperture diameter. This physical limit explains why continuing to reduce pupil size can eventually impair fine detail even after geometric blur has decreased.

Contrast transfer is a curve, not one number

Equal cutoff frequencies do not imply equal contrast at all lower spatial frequencies.

Original comparison

Spatial frequency	System A MTF	System B MTF
10 units	0.8	0.6
30 units	0.5	0.2
Cutoff	Same	Same

Read the consequence

For the same input modulation at 30 units, system A transfers more modulation than system B. Both may have the same highest transmitted frequency while differing in useful contrast over the remaining band. The illustrative values describe hypothetical systems, not a particular clinical instrument.

Point spread and modulation transfer describe one system

A point image and a sinusoidal-pattern response are complementary optical descriptions.

Link the measurements

- An ideal point is imaged as a point-spread function
↓
- Its spatial distribution describes blur
↓
- Optical transfer expresses spatial-frequency response
↓
- MTF is the magnitude of that transfer

The limit

MTF does not preserve phase information on its own. Two systems can therefore share an MTF yet differ in aspects of image phase. Specify whether the comparison concerns contrast amplitude, phase or the complete optical transfer function.

Acuity samples only one part of visual performance

A high-contrast letter endpoint can miss losses at lower contrast.

Readout

Feature	Question answered	Not established
High-contrast acuity	Smallest recognised detail	All contrast sensitivity
Contrast threshold	Detectability at selected contrast/frequency	Anatomical localisation alone
Optical MTF	Transferred modulation	Entire neural performance

A comparison experiment

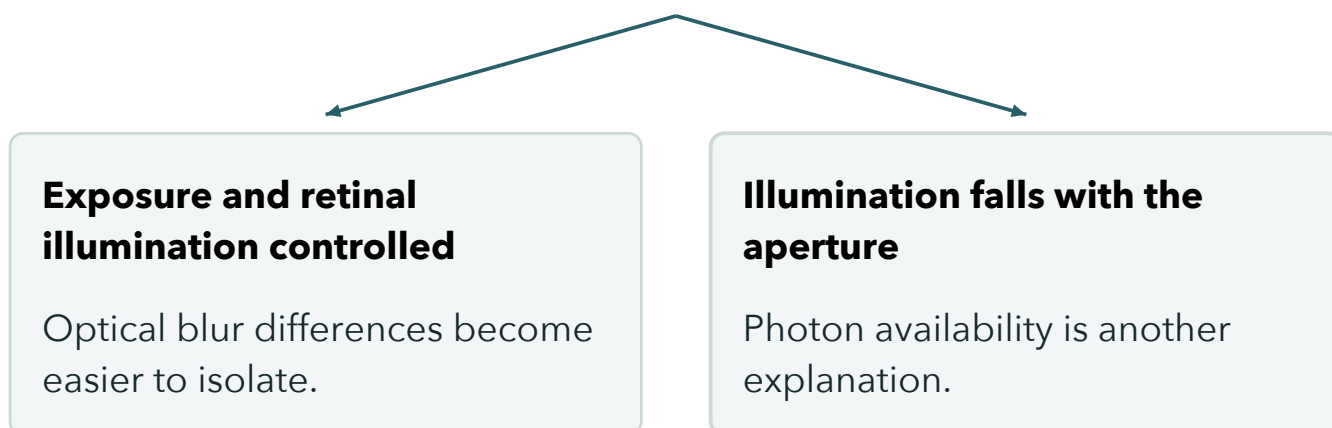
Keep retinal luminance and task conditions comparable when studying pupil effects. A change in visibility may reflect optical transfer, photon availability, neural processing or several of these. An optical calculation alone is not a full prediction of perception.

Choose the relevant scale

Resolution, brightness and contrast are related but distinct endpoints.

Two apertures give different performance

What is held fixed?



Postgraduate reasoning

State the wavelength, aperture and optical assumptions before applying a diffraction formula. A geometrical ray diagram is useful for aberrations but cannot alone describe a wave-optical diffraction limit. Combining both models explains why the preferred aperture need not be the smallest available one.